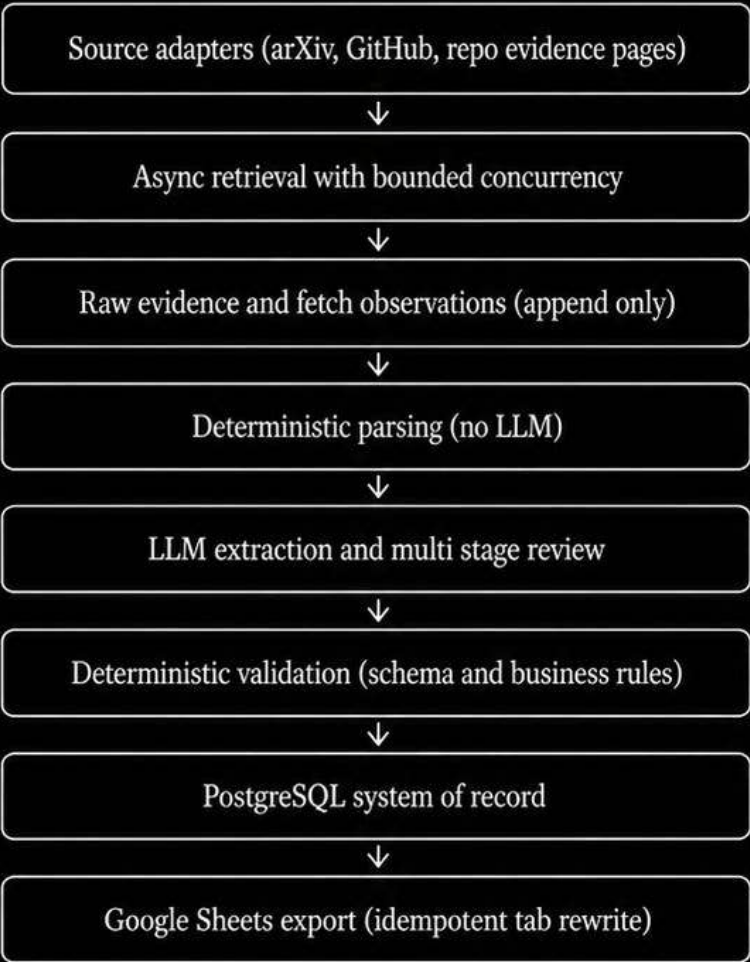


GraphOneSlice: Production Data Intelligence Architecture

SYSTEM ARCHITECTURE AND END TO END DATA FLOW



EVIDENCE AND PROVENANCE MODEL

Every stage writes what it observed before any stage writes what it concluded. Raw arXiv Atom responses, abstract page HTML, and GitHub API snapshots are persisted as evidence rows, not discarded after parsing.

A source observation and the record extracted from it are not assumed to be one to one. The arXiv discovery fetch is linked to papers through a join table, because one Atom page can supply many papers. The abstract page HTML fetch is linked through a scalar paper id, because that fetch is genuinely one to one with a single paper.

This lineage is what lets a reviewer trace any exported field back to the exact source response that produced it.

STORAGE AND VALIDATION

PostgreSQL is the single system of record. Fetch observations, repository link candidates, GitHub snapshots, and validated records are append only or idempotently keyed, so a rerun never silently duplicates evidence.

A record only becomes exportable after it passes deterministic schema and business rule validation. Model output is never exported on its own authority.

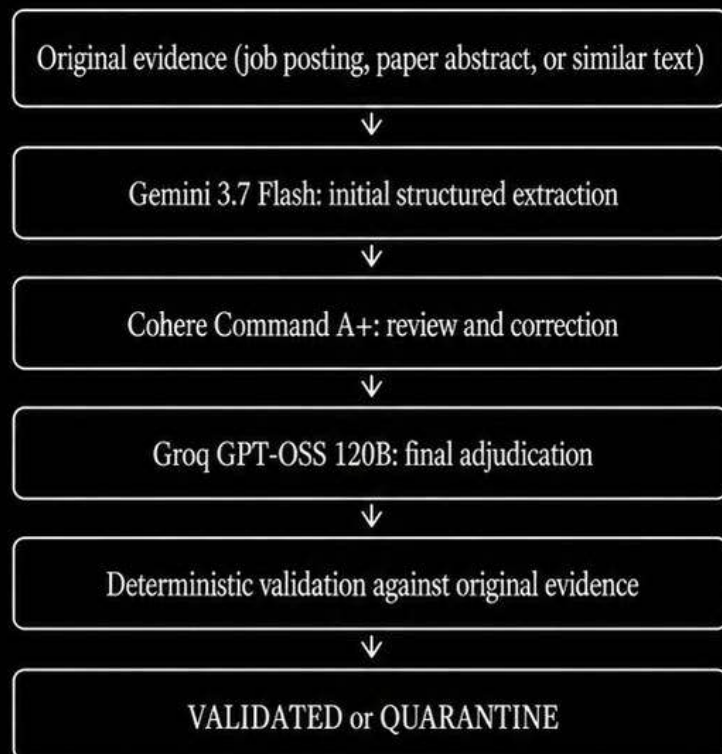
Google Sheets is a downstream export sink, not a source of record. Export is a clear and rewrite of the target tab, which keeps a rerun after a partial failure safe by construction rather than by convention.

SOURCES REPRESENTED IN THE PROVEN PATH

- arXiv: paginated Atom discovery feed, plus the per paper abstract page HTML fetched as separate evidence.
- GitHub API: repository existence verification and live stargazers_count observation per resolved paper.
- PostgreSQL: papers, fetch_observations, paper_repo_links, github_repo_snapshots, validated_records, pipeline_log.
- Google Sheets: ResearchPapers tab, written from validated_records only, never written to directly by any earlier stage.

LLM Orchestration, Resilience, and Scale Strategy

EVIDENCE GROUNDED EXTRACTION CHAIN



DESIGN PRINCIPLES

The original evidence is authoritative throughout the chain. A later provider may correct or add a field only when the original evidence supports that change.

Agreement between models is never treated as evidence. Each extraction claim must carry a quote that is checked against the source text before the record can be validated.

A fallback execution is recorded distinctly from a normal execution, so an audit trail never describes a fallback extraction as if it were an ordinary review or adjudication step.

FAILURE HANDLING

HTTP 429: computed exponential backoff with jitter, honoring a provider Retry-After header when present, across a bounded number of attempts before the call is treated as failed.

HTTP 413: classified as a distinct payload too large failure rather than retried blindly. Bounded chunking of oversized evidence is part of the scale design; it is not yet wired into the live provider call path, which currently surfaces 413 as an error to the caller.

Provider isolation: a Gemini failure triggers a defined Cohere fallback extraction, followed by Groq adjudication, rather than aborting the record.

500,000 RECORD SCALE ARCHITECTURE (DESIGN TARGET, NOT BENCHMARKED)

- Asynchronous I/O with bounded concurrency per source, so throughput scales without changing application logic.
- Work is partitionable by source and by page, so additional workers claim unclaimed batches rather than requiring a rewrite.
- PostgreSQL remains the single system of record; idempotent state transitions make a rerun safe after a crash or restart.
- Append only fetch and verification observations, plus content and source hashing, avoid silent duplication as volume grows.
- Retry and quarantine queues isolate a bad record from the batch instead of failing the whole run.
- Horizontal worker scaling and downstream export sinks (Sheets, or a warehouse later) are additive, not architectural rewrites.

Implementation Evidence, Status, and Production Gaps

ASSESSMENT AREA STATUS

ASSESSMENT AREA	STATUS
Research papers (1,000 records)	LIVE PROVEN
GitHub repository verification and stars	LIVE PROVEN
PostgreSQL persistence	LIVE PROVEN
Google Sheets export (ResearchPapers tab)	LIVE PROVEN
Automated test suite (55 of 55)	LIVE PROVEN
LLM orchestration (Gemini, Cohere, Groq)	IMPLEMENTED
LLM fallback path (Cohere extract, Groq adjudicate)	LIVE TESTED
Normal Gemini to Cohere to Groq path	NOT FULLY LIVE PROVEN
Startups (target 1,000+)	NOT COMPLETED
Products (target 1,000+)	NOT COMPLETED
News (5 sources, 24 hour freshness)	NOT COMPLETED
Jobs (5 boards, 24 hour freshness)	NOT COMPLETED
Entity resolution and mapping log	DESIGNED, NOT COMPLETE
Anti bot / Cloudflare / Datadome strategy	DESIGN, NOT IMPLEMENTED
500,000 record benchmark	NOT RUN

KNOWN IMPLEMENTATION BOUNDARIES

- arXiv discovery currently logs and stops on a fetch exception rather than performing sophisticated automatic page resume.
- The GitHub client computes backoff and jitter; its Retry-After header callback is a placeholder and does not yet parse the header value.
- An abstract page fetch failure can be recorded as resolved with no repository found, which is not the same as proving no repository exists.
- Repository link candidates are not guaranteed to be perfectly deduplicated under every evidence condition.
- The validator does not guarantee that a claim value is byte for byte identical to the extraction field value it supports.
- Sheets export is a clear then write operation; it is idempotent on rerun but not transactional mid write.

PRODUCTION NEXT STEPS

- Complete the Startup, Product, News, and Jobs source adapters against the 1,000+ record and 24 hour freshness targets.
- Wire chunked and truncated evidence submission into the live provider call path for HTTP 413.
- Complete deterministic entity resolution and produce the entity mapping log.
- Implement production anti bot handling for JavaScript heavy and bot protected sources.
- Add distributed work claiming for horizontal worker scaling beyond a single process.
- Parse Retry-After and X-RateLimit-Reset headers directly instead of relying on computed backoff alone.

Every extracted claim must remain traceable to legitimate source evidence. The system treats evidence as authoritative and quarantines records that cannot be deterministically validated.